

Lesson 1: Ohm's Law and Basic Circuits

Learning objectives

- State Ohm's law and use it to find voltage, current or resistance.
- Calculate total resistance in series and parallel circuits.

1.1 Voltage, current and resistance

Voltage (V) is the electrical push that drives charge round a circuit and is measured in volts. **Current (I)** is the flow of charge, measured in amperes. **Resistance (R)** opposes the flow of current and is measured in ohms.

1.2 Ohm's law

For a fixed resistor, the current is directly proportional to the voltage: $V = I \times R$. Rearranged, $I = V / R$ and $R = V / I$.

Worked example: A 12 V battery is connected across a 6 ohm resistor. Current = $12 / 6 = 2$ A.

1.3 Series and parallel

In a **series** circuit the same current flows through every component and the resistances add: $R_{\text{total}} = R_1 + R_2 + R_3$.

In a **parallel** circuit each branch has the same voltage and the currents add. For two resistors, $R_{\text{total}} = (R_1 \times R_2) / (R_1 + R_2)$.

Quantity	Symbol	Unit	Measured with
Voltage	V	Volt (V)	Voltmeter (in parallel)
Current	I	Ampere (A)	Ammeter (in series)
Resistance	R	Ohm	Ohmmeter or multimeter
Power	P	Watt (W)	$P = V \times I$

Key points

- $V = I \times R$.
- Series: same current, resistances add.
- Parallel: same voltage, total resistance is smaller than the smallest branch.

Review questions

- 1 A 9 V supply drives a current of 3 A. Find the resistance.
- 2 Find the total resistance of 10, 20 and 30 ohm resistors in series.
- 3 Why is an ammeter connected in series?

Lesson 2: Common Electronic Components

Learning objectives

- Identify common components and their symbols' purposes.
- Describe the job of a resistor, capacitor, diode, LED and transistor.

2.1 Passive and active components

Passive components (resistors, capacitors) do not amplify or switch a signal. Active components (diodes, transistors) control the flow of current and can switch or amplify.

2.2 Component notes

Resistor: limits current. Its value is shown by coloured bands.

Capacitor: stores electric charge and smooths voltage. Value is measured in farads (usually microfarads).

Diode: lets current flow in one direction only. Used in rectifiers that change AC to DC.

LED: a diode that gives out light. It always needs a resistor in series to limit the current.

Transistor: works as an electronic switch or amplifier. A small base current controls a larger collector current.

Component	Function	Where you find it
Resistor	Limits current	Almost every circuit
Capacitor	Stores charge, filters ripple	Power supplies, timers
Diode	One-way current flow	Chargers, rectifiers
LED	Emits light	Indicators, lamps
Transistor	Switch or amplifier	Amplifiers, logic circuits

Key points

- A diode conducts in one direction only.
- An LED must have a series resistor.
- A transistor can act as a switch.

Review questions

- 1 What is the function of a capacitor?
- 2 Why must an LED be used with a series resistor?
- 3 Name the component used to change AC into DC.